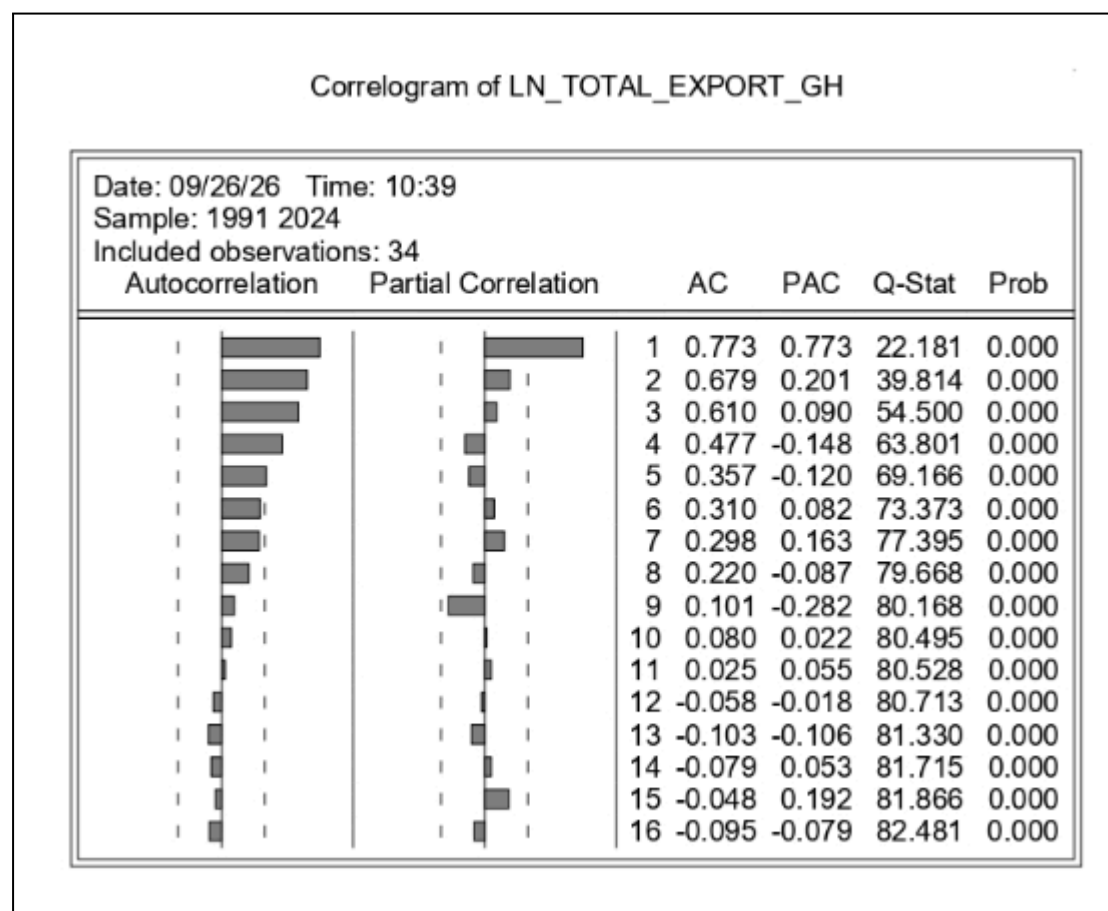
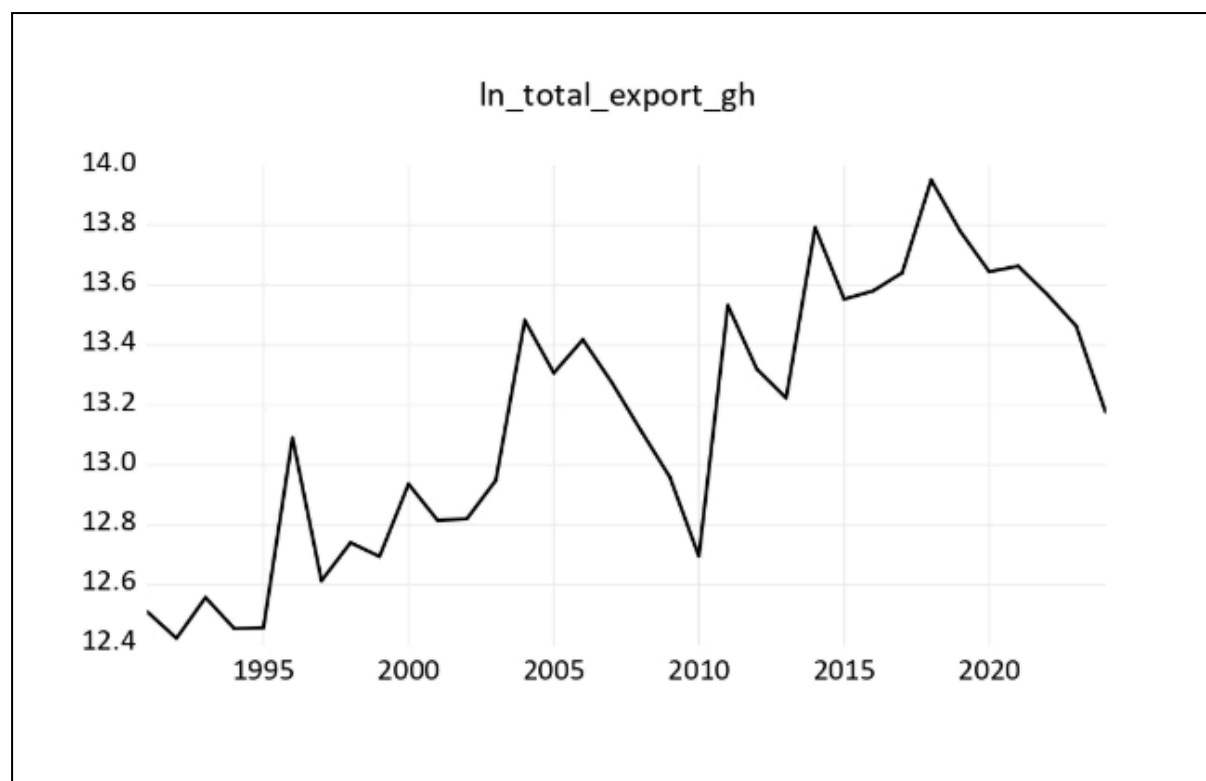


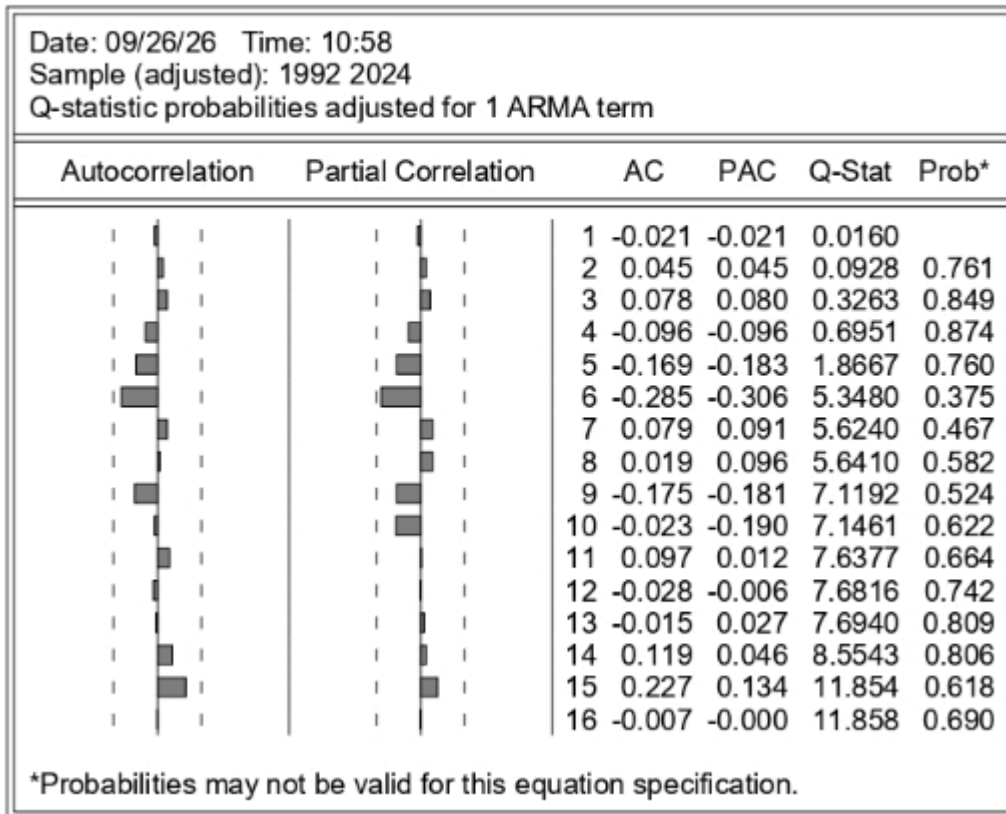


Debido a que la Función de Autocorrelaciones tarda varios desfases en anularse, y la Función de Autocorrelaciones Parciales se anula abruptamente en el segundo desfase,

estimo el modelo como un AR(1), con coeficiente y con tendencia (positiva) por como se ve el gráfico

Dependent Variable: LN_TOTAL_EXPORT_GH				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/26/26 Time: 10:40				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Convergence achieved after 2 iterations				
Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	12.55200	0.135572	92.58538	0.0000
@TREND	0.036049	0.006906	5.220176	0.0000
AR(1)	0.326757	0.195394	1.672295	0.1049
R-squared	0.706174	Mean dependent var	13.17251	
Adjusted R-squared	0.686586	S.D. dependent var	0.443150	
S.E. of regression	0.248091	Akaike info criterion	0.136462	
Sum squared resid	1.846468	Schwarz criterion	0.272508	
Log likelihood	0.748375	Hannan-Quinn criter.	0.182238	
F-statistic	36.05065	Durbin-Watson stat	1.901546	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.33			

Correlogram of Residuals



Se ve como la tendencia con coeficiente positivo, y el coeficiente (ϕ_0) dieron significativos, sin embargo el coeficiente del AR(1) (ϕ_1) no dio significativo al 10%.

Dependent Variable: LN TOTAL EXPORT GH

Method: Least Squares

Date: 09/26/26 Time: 19:49

Sample: 1991 2024

Included observations: 34

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	12.52824	0.084289	148.6342	0.0000
@TREND	0.037865	0.004391	8.623567	0.0000
R-squared	0.699152	Mean dependent var	13.15301	
Adjusted R-squared	0.689750	S.D. dependent var	0.450959	
S.E. of regression	0.251184	Akaike info criterion	0.131762	
Sum squared resid	2.018993	Schwarz criterion	0.221548	
Log likelihood	-0.239962	Hannan-Quinn criter.	0.162382	
F-statistic	74.36591	Durbin-Watson stat	1.299920	
Prob(F-statistic)	0.000000			

Date: 09/26/26 Time: 19:50

Sample: 1991 2024

Included observations: 34

Included observations: 64						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	0.261	0.261	2.5177	0.113
		2	0.097	0.032	2.8810	0.237
		3	0.029	-0.004	2.9145	0.405
		4	-0.163	-0.185	4.0012	0.406
		5	-0.299	-0.239	7.7830	0.169
		6	-0.325	-0.215	12.398	0.054
		7	-0.008	0.171	12.401	0.088
		8	-0.023	-0.018	12.425	0.133
		9	-0.154	-0.254	13.589	0.138
		10	-0.027	-0.136	13.626	0.191
		11	0.125	0.101	14.452	0.209
		12	0.021	-0.005	14.478	0.271
		13	0.048	0.035	14.612	0.332
		14	0.214	0.089	17.423	0.234
		15	0.199	0.041	19.976	0.173
		16	-0.026	-0.090	20.021	0.219

Eliminando el componente AR(1) tanto la constante como la tendencia siguen siendo significativos.

Y al analizar los residuos vemos que se comportan como ruido blanco, lo que podría indicar que tiene tendencia determinística.

Sin embargo hay que realizar el test de Dickey-Fuller Ampliada para descartar la posibilidad de que tenga raíz unitaria.

Null Hypothesis: LN TOTAL EXPORT GH has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-3.445559	0.0625
Test critical values:				
	1% level		-4.262735	
	5% level		-3.552973	
	10% level		-3.209642	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN TOTAL EXPORT GH)				
Method: Least Squares				
Date: 09/26/26 Time: 19:47				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN TOTAL EXPORT GH(-1)	-0.673243	0.195394	-3.445559	0.0017
C	8.462324	2.434370	3.476186	0.0016
@TREND("1991")	0.024270	0.009253	2.622779	0.0136
R-squared	0.293695	Mean dependent var		0.020227
Adjusted R-squared	0.246608	S.D. dependent var		0.285825
S.E. of regression	0.248091	Akaike info criterion		0.136462
Sum squared resid	1.846468	Schwarz criterion		0.272508
Log likelihood	0.748375	Hannan-Quinn criter.		0.182238
F-statistic	6.237285	Durbin-Watson stat		1.901546
Prob(F-statistic)	0.005431			

Null Hypothesis: LN TOTAL EXPORT GH has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-2.168777	0.2208
Test critical values:				
	1% level		-3.646342	
	5% level		-2.954021	
	10% level		-2.615817	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN TOTAL EXPORT GH)				
Method: Least Squares				
Date: 09/26/26 Time: 19:48				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN TOTAL EXPORT GH(-1)	-0.226547	0.104459	-2.168777	0.0379
C	2.999843	1.374677	2.182217	0.0368
R-squared	0.131740	Mean dependent var		0.020227
Adjusted R-squared	0.103732	S.D. dependent var		0.285825
S.E. of regression	0.270594	Akaike info criterion		0.282300
Sum squared resid	2.269861	Schwarz criterion		0.372998
Log likelihood	-2.657952	Hannan-Quinn criter.		0.312817
F-statistic	4.703592	Durbin-Watson stat		2.473875
Prob(F-statistic)	0.037890			

El primer ADF test analiza un modelo con tendencia y constante, y tiene un p-valor de 0,0625 por lo que con un nivel de significancia del 5% no se puede rechazar la hipótesis

nula de raíz unitaria.

El segundo test, con constante y sin tendencia arroja un p-valor más alto (0,2208), por lo que ahora no se rechaza la hipótesis nula ni con un nivel de significancia del 10% por lo que en ninguno se afirmar que el modelo no tenga raíz unitaria, por lo tanto es necesario diferenciar una vez la variable.

$\Delta(\ln_total_export_gh)$

Date: 09/26/26 Time: 19:50		Sample (adjusted): 1992 2024		Included observations: 33 after adjustments	
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.372	-0.372	5.0054	0.025
		2 -0.040	-0.208	5.0660	0.079
		3 0.126	0.037	5.6782	0.128
		4 -0.038	0.029	5.7361	0.220
		5 -0.043	-0.029	5.8123	0.325
		6 -0.217	-0.315	7.8341	0.251
		7 0.195	-0.037	9.5199	0.217
		8 0.085	0.177	9.8539	0.275
		9 -0.197	-0.024	11.717	0.230
		10 0.046	-0.116	11.822	0.297
		11 0.073	-0.059	12.103	0.356
		12 -0.070	-0.058	12.372	0.416
		13 -0.050	-0.002	12.516	0.486
		14 -0.052	-0.111	12.681	0.552
		15 0.226	0.092	15.963	0.385
		16 -0.012	0.182	15.973	0.455

Se ve como tanto en la FAC como FACP solo el primer rezago es significativo por lo que podría ser un ARIMA(1,1,1).

Dependent Variable: D(LN TOTAL EXPORT GH)
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 09/26/26 Time: 19:54
Sample (adjusted): 1993 2024
Included observations: 32 after adjustments
Failure to improve likelihood (non-zero gradients) after 16 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1992

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.034451	0.011084	3.108003	0.0042
AR(1)	0.322743	0.209284	1.542130	0.1339
MA(1)	-0.928009	0.079549	-11.66582	0.0000
R-squared	0.248649	Mean dependent var		0.023628
Adjusted R-squared	0.196831	S.D. dependent var		0.289719
S.E. of regression	0.259645	Akaike info criterion		0.230060
Sum squared resid	1.955056	Schwarz criterion		0.367473
Log likelihood	-0.680958	Hannan-Quinn criter.		0.275608
F-statistic	4.798563	Durbin-Watson stat		1.901522
Prob(F-statistic)	0.015839			
Inverted AR Roots	.32			
Inverted MA Roots	.93			

El coeficiente ϕ_1 del componente AR no es significativo por lo que lo retiro de la estimación.

Dependent Variable: D(LN TOTAL EXPORT GH)
 Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
 Date: 09/26/26 Time: 19:55
 Sample (adjusted): 1992 2024
 Included observations: 33 after adjustments
 Failure to improve likelihood (non-zero gradients) after 12 iterations
 Coefficient covariance computed using outer product of gradients
 MA Backcast: 1991

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.028921	0.022108	1.308168	0.2004
MA(1)	-0.540173	0.170880	-3.161116	0.0035
R-squared	0.188953	Mean dependent var		0.020227
Adjusted R-squared	0.162790	S.D. dependent var		0.285825
S.E. of regression	0.261527	Akaike info criterion		0.214135
Sum squared resid	2.120292	Schwarz criterion		0.304833
Log likelihood	-1.533234	Hannan-Quinn criter.		0.244652
F-statistic	7.222182	Durbin-Watson stat		1.842635
Prob(F-statistic)	0.011475			
Inverted MA Roots	.54			

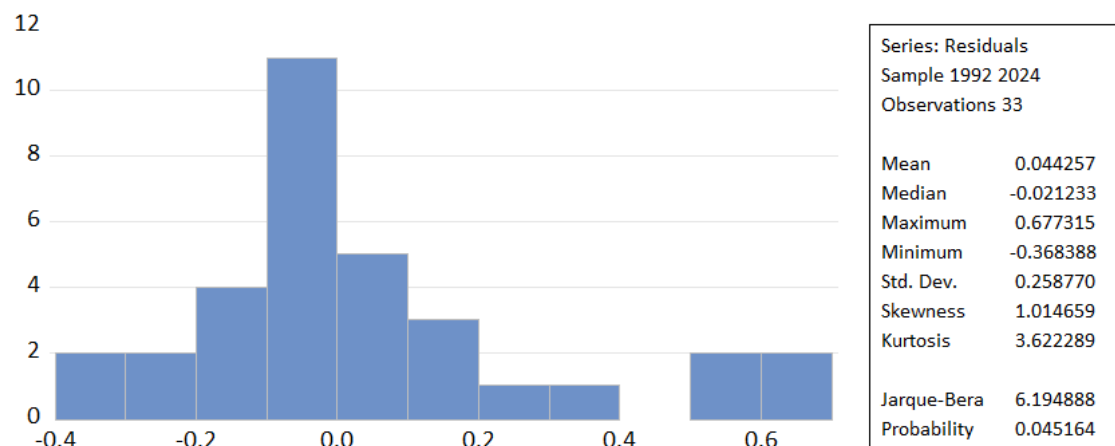
Al sacar el componente AR la constante dejó de ser significativa por lo que también la retiro.

Dependent Variable: D(LN TOTAL EXPORT GH) Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps) Date: 09/26/26 Time: 19:56 Sample (adjusted): 1992 2024 Included observations: 33 after adjustments Failure to improve likelihood (non-zero gradients) after 7 iterations Coefficient covariance computed using outer product of gradients MA Backcast: 1991				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	-0.433615	0.167110	-2.594789	0.0142
R-squared	0.155625	Mean dependent var		0.020227
Adjusted R-squared	0.155625	S.D. dependent var		0.285825
S.E. of regression	0.262644	Akaike info criterion		0.193800
Sum squared resid	2.207420	Schwarz criterion		0.239149
Log likelihood	-2.197697	Hannan-Quinn criter.		0.209058
Durbin-Watson stat	1.969187			
Inverted MA Roots	.43			

El único coeficiente restante es significativo al 5% por lo que verifico los residuos.

Date: 09/26/26 Time: 19:56						
Sample (adjusted): 1992 2024						
Q-statistic probabilities adjusted for 1 ARMA term						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	-0.055	-0.055	0.1095	
		2	-0.042	-0.045	0.1757	0.675
		3	0.077	0.072	0.4032	0.817
		4	-0.077	-0.072	0.6415	0.887
		5	-0.169	-0.173	1.8191	0.769
		6	-0.241	-0.285	4.3104	0.506
		7	0.161	0.130	5.4673	0.485
		8	0.083	0.122	5.7892	0.565
		9	-0.173	-0.160	7.2335	0.512
		10	-0.006	-0.156	7.2352	0.613
		11	0.080	-0.014	7.5719	0.671
		12	-0.079	-0.036	7.9195	0.720
		13	-0.071	0.008	8.2079	0.769
		14	0.066	-0.000	8.4746	0.811
		15	0.245	0.159	12.322	0.580
		16	0.017	0.077	12.340	0.653

Los residuos del modelo ARIMA(0,1,1) sin constante se comportan como ruido blanco. Ahora verificamos con el test de normalidad.



Con un nivel de significancia del 5% el test de Jarque-Bera confirma que los residuos se comportan de forma Normal.

De este modo queda modelado la variable $\ln_total_export_gh$, que representa la oferta de cacao de Ghana, formada por la exportacion de cacao y de todos sus derivados debidamente ponderados, como un ARIMA(0,1,1).

